



Huge artillery guns such as the Howitzer cannon were transported by horses and tractors. High-explosive shells fired in massive quantities were key to the war's high casualty rates.

Tannenberg, but its Austrian allies suffered several defeats. On the Western Front, however, a 400-mile (645-km) trench line stretched from the Belgian coast in the north, down through eastern France to the Swiss border. The two sides faced each other across the open space between their front lines. This area—no-man's-land—had barbed wire fronting the trenches to slow the opposition. Continuous fighting from the trenches, punctuated by appallingly bloody battles, failed to break the deadlock. More than 600,000 Allied troops were killed or wounded in the Battle of the Somme alone.

Total war

At the start of the conflict, both sides had been convinced it would be a short, decisive battle. No one had anticipated a war of attrition. New mechanized weapons added to the high casualty rates. Tanks were used for the first time, and machine guns such

as the German MG 08 Maxim could fire up to 600 bullets a minute. Aircraft, first employed for reconnaissance, were later used for bombing. Both sides used poison gas. Horses were the backbone of logistical operations, but as the war progressed, railroads and motor trucks were used to transport goods to the front.

Civilians were brought into the front line by the bombing of London and Paris by airships and bomber aircraft. By 1917, German submarines were sinking one in four merchant ships headed for Britain to try to starve the British into submission. Britain's naval blockade of Germany also led to acute food shortages. This was the first "total war," meaning not just soldiers but also civilian populations were involved.

Britain was forced to introduce conscription for the first time in its history. From January 1916, all single men aged 18 to 41 could be called up. Britain and France also assembled armies from their overseas colonies, such as India and Africa, and from the British dominions of Australia, New Zealand and Canada. War brought many social changes, notably for women, who filled positions in factories and offices. Women were also increasingly employed in the munitions industry as governments switched to large-scale production.

Life in the trenches



At the outbreak of World War I, both sides anticipated fast-moving battles that would cover hundreds of miles. None expected a static fight

with their forces deep in defensive trenches.

Early trenches were small furrows, but they grew more elaborate, fortified with wooden frames and sandbags. German trenches were more sophisticated and had electricity and toilets. Soldiers spent daylight hours avoiding enemy fire and endured periods of boredom and daily chores, broken up with spells in reserve and short rest periods.

The trenches sometimes filled up with

rats and lice, as well as water, which turned to ice. Life in such conditions was exhausting, and soldiers had a repetitive diet of canned food and few comforts.

Snipers shot at any heads that appeared over the parapet, and raiding parties throwing grenades were a constant danger. The trenches were bombarded with shells, bullets, and poisonous gas. It was a relentless war of attrition—smelly, dirty, and riddled with disease.